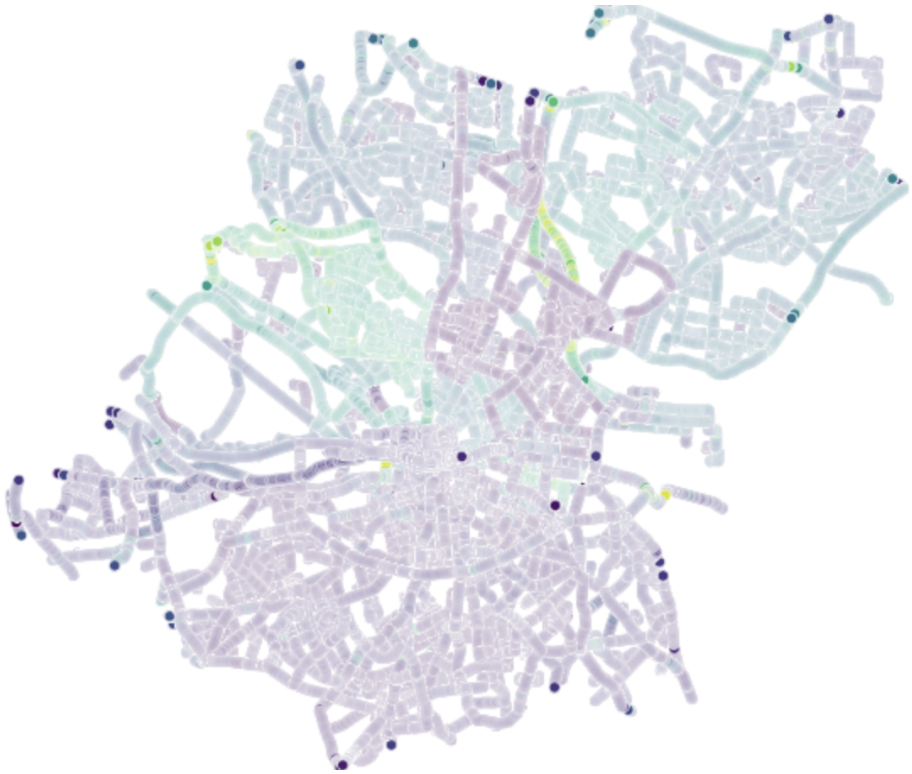
Forecasting Next-Day PM2.5 in Dublin Using Mobile Air Quality, Weather, and Traffic Data

Intro to Machine Learning Group Project



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1. Objective and motivation

This project investigates what factors impact air pollution in Dublin and, specifically, to develop a forecasting model capable of accurately predicting air pollution for the next day or days. The project is not solely about predictive accuracy; it asks whether a usable, policy-relevant next-day signal can be extracted from the available data.

We focused on forecasting PM2.5 particles, a pollutant identified by world health organisations as one of the most dangerous to public health [6]. PM2.5 refers to particulate matter smaller than 2.5 micrometres in diameter [1]. It has been found to decrease visibility, penetrate the lungs, and cause harm to the respiratory, cardiovascular, cerebrovascular, and nervous systems [1]. It is also closely related to mortality [1].

The practical aim of this work is to inform monitoring, communication, and intervention timing for DCC to help improve air quality in Dublin [7]. DCC should therefore treat the results of this project as a proof of concept for short-term PM2.5 forecasting, one that can guide decisions about where to invest in better data infrastructure and operational tools.

2. Data

Three data sources underpin this project: Google Air View mobile pollution measurements, Met Éireann weather records, and SCATS traffic volumes. PM2.5 is the target pollutant, weather captures the main naturogenic driver of particulate dispersion, and traffic captures the principal anthropogenic emission source in an urban setting. Meteorological conditions, emissions, and external transport have been found to influence atmospheric particulate matter concentrations [1], while in urban areas the main human-generated sources include exhaust emissions, road surface abrasion, brake and tyre wear, and particle resuspension from paved roadways [2].

Google Air View data - The core dataset consists of street-view car pollutant measurements recorded at one-second intervals across a 15-month period during working hours (May 2021–August 2022), containing over five million timestamped mobile observations linked to GPS coordinates. The data reflect daytime mobile observations taken across Dublin city, not a fixed-station monitoring system. The dataset offers unusually rich spatial detail and street-level readings representative of individual exposure, but coverage is mobile, uneven, and weekday-only, with daily observation counts varying substantially. Rush-hour data (before 9 am and after 6 pm) and weekend or public holiday observations are absent. The data are therefore sufficient for a city-wide next-observed-weekday forecasting exercise, but not for a full seven-day, 24-hour ambient system, and are better suited to a Dublin-wide daily average target than to highly local forecasting.

Weather and traffic data - Weather data were taken from Met Éireann's Dublin Airport daily series [4], including windspeed, evaporation measures (a proxy for humidity), and temperature—all found to be significant predictors of PM2.5 [1]. Traffic data come from the Sydney Coordinated Adaptive Traffic System (SCATS), which uses sensors at each traffic signal to detect vehicle presence [5]. The dataset provides total and average traffic volumes per preceding hour. Data were initially available at region-day level and were aligned to the Dublin-wide target through spatial aggregation; a North/South Dublin version was created later as a robustness check.

3. Factor choice

Data preparation - During visualisation the data were explored to reveal coverage restrictions, relationships between variables, descriptive statistics, spread, granularity, and missingness. The Air View data showed that weekends were not present and that there were fewer recordings on Mondays and Fridays. For preprocessing, any PM2.5 value above 400 was removed after consultation with a news article indicating that such readings could be reported as anomalies [3]. From the weather data only columns potentially correlated with PM2.5 were retained, and from the traffic data only the mean, median, and total daily site volumes were kept.

All data were aggregated by location using averages across Dublin for PM2.5 and traffic measurements; weather was recorded at only one location (Dublin Airport). This was a pragmatic decision to align the observational unit across all three data sources. It makes the modelling problem tractable and defensible given the available weather and traffic data, and it means the results speak to city-wide daily conditions rather than local hotspots. All datasets were then merged by date, including variables for day of the week, day name, and the averaged, relevant pollution, weather, and traffic data.

Weekends are structurally out of scope rather than random missing observations, and even after accounting for that, the dataset still contains some unexpected missing weekdays. Coverage diagnostics suggest the target is reasonably stable, but uneven observation counts remain a real limitation.

Exogenous variable selection - Predictor selection aimed to keep the final model parsimonious and interpretable. Redundant or overlapping variables were removed to reduce noise and unnecessary complexity, which was especially important given the modest city-day sample size. Correlation matrices between traffic variables and PM2.5 values and between weather variables and PM2.5 values were used to guide selection.

For traffic, Dublin mean traffic values were chosen because the mean had a larger spread of values than the median and total. The final traffic variable used was `dublin_mean_site_traffic`.

For weather, three variables were dropped. Rainfall did not appear to be correlated with PM2.5 levels at all. Potential evapotranspiration (`pe`) was highly correlated with evaporation and is conceptually very similar. The highest 10-minute windspeed (`hm`) could be dropped because other, more influential wind measures were available. The final weather variables retained were: `maxtp` (maximum temperature), `mintp` (minimum temperature), `wdsp` (mean wind speed in knots), `hg` (highest gust in knots), `sun` (sunshine duration in hours), `g_rad` (global radiation in J/cm^2), and `evap` (evaporation in mm).

Temporal feature selection - No strong seasonality or trend was noted throughout the time period. Daily averages appear to change slightly by weekday, indicating that the `day_of_week` variable is beneficial to include in the model. Temporal diagnostics support keeping a short lag structure, with `lag_1` being the best predictor and lags 2 and 3 showing less potential predictive power but enough to be included. The final temporal features were: `day_of_week`, `lag_1`, `lag_2`, and `lag_3`.

4. Method

Train/test split - A single shared chronological train–test split was created on the observed rows. The split is designed to preserve the forward-looking forecasting setting, so all models are evaluated on a genuinely held-out future period. This avoids leakage from future observations into training. All subsequent models draw from this split and then apply any model-specific preprocessing afterwards, ensuring accurate and consistent comparisons between models.

Baseline model - The most basic model was used to obtain evaluation metrics against which more complex models could be compared. This baseline predicts each observed test-day PM2.5 value using the previous observed PM2.5 value; the first test prediction therefore uses the last observed training value.

Full models

Weekday Average - This model took the average of the last four values for a given weekday and used it as the prediction for all future occurrences of that weekday. It served as a second baseline to determine how well averaged past values, especially those grouped by weekday, performed in the forecasting task.

Linear Regression - This model used the past three days of data from all variables to create a weighted, linear combination of these predictors. It is fairly simple and easy to interpret, showing how PM2.5 values change as a function of the selected predictors.

Direct One-Step-Ahead Ridge ARX - This is a suitable full model for the project because the task is one-step-ahead prediction, the sample is relatively small, and several retained weather variables remain correlated. Ridge regularisation shrinks unstable coefficients without sacrificing interpretability, making it a sensible ARX specification for a compact city-day dataset. A key principle is that the model uses only information that would actually be available at forecast time: weather and traffic therefore enter in lagged form rather than as tomorrow's realised values. This makes the forecasting exercise more realistic but also more demanding.

Random Forest - Random forests are beneficial for small datasets and are highly interpretable. This model provides a fairly different approach from the linear and ridge regressions, helping to assess whether an entirely different forecasting paradigm would be beneficial. It was included mainly as a contrasting non-linear benchmark against the linear and ridge specifications, to determine whether a more flexible model adds value beyond the simpler forecasting approaches.

5. Results

The models were compared using root mean squared error (RMSE) on the held-out test set. The results are summarised below:

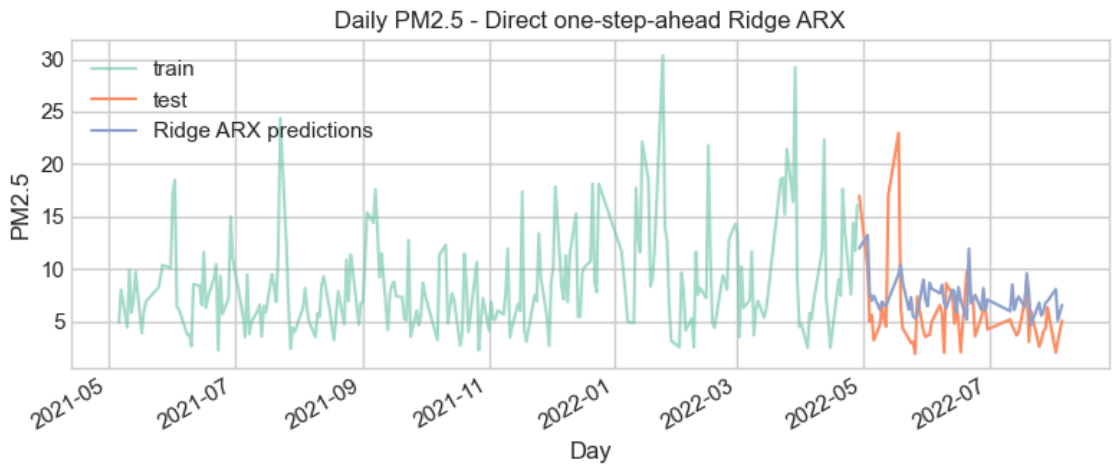
Table 1: RMSE comparison across models.

Model	**RMSE**
Baseline (persistence)	3.86
Weekday average	5.92
Linear regression	3.94
Direct one-step-ahead Ridge ARX	3.72
Random forest	4.62

The Direct one-step-ahead Ridge ARX appeared to give the most accurate predictions, providing

a promising ability to forecast PM2.5 for the next day. The improvement over the simple persistence baseline is real but modest, which suggests that most short-term predictability comes from recent PM2.5 history. Weather and traffic add incremental signal but do not transform forecast accuracy.

Among the other models, the same-weekday average model performed poorly, suggesting that a simple weekly cycle is not enough to forecast PM2.5 well. The random forest did not outperform the more interpretable linear-style models. Forecast gains therefore came more from careful feature design than from model complexity.



6. Discussion and recommendation

Main findings - We were able to accurately predict next-day PM2.5 using the data, features, processing, and model tuning employed in this project. The strongest and most defensible predictive signal is temporal persistence, especially yesterday's PM2.5. The project therefore supports a light-touch next-day decision-support tool; it does not yet support a highly accurate standalone operational forecasting system.

Limitations - The weather data may not be the most accurate given the data sources available. With weather data recorded at a location closer to or within the city centre, PM2.5 predictions within that area could be more accurate. Weekend data would provide more continuous and generalisable observations to work with, and better location indicators would allow disaggregation of the data for location-specific forecasts.

The All-Dublin aggregation is sensible for the main model, but it necessarily masks neighbourhood-level variation. The North/South Dublin split was explored as a robustness check. The limited gains from weather and traffic should not be read as evidence that those factors are unimportant; their predictive value is constrained by using only information available at forecast time $(t-1)$.

Recommendations - DCC should treat this as a proof of concept for short-term PM2.5 forecasting. The most valuable next improvements are likely to come from better temporal continuity, more spatially balanced monitoring, more local weather inputs, and forecast weather and forward-looking traffic inputs. Better data design is likely to yield more policy value than simply moving to more complex algorithms.

6. References

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